Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Backend software engineer with 2+ years building **Java/Spring Boot services**, REST APIs, microservices, and financial workflows; delivered tested distributed systems handling 2M+ records and 1M+ transactions across AWS with strong data integrity.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed **backend services and normalized relational schemas** for 500+ users, implementing row-level isolation, indexed queries, and reusable access patterns that reduced API latency from 1.2 seconds to 350 milliseconds.
- Maintained **distributed data integrity** across 5,000+ monthly jobs by implementing explicit states, validation, reconciliation, idempotent processing, and failure isolation, achieving 99%+ production reliability through documented CI/CD releases.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **Python Flask APIs and backend services** handling 10,000+ daily requests for 150+ users, applying multithreading, secure validation, persistent results, and fault-tolerant execution across 1,000+ files per batch.
- Collaborated through **sprint planning, code reviews, testing, and technical discussions**, debugging production integrations and deploying GitHub Actions-backed enhancements that reduced turnaround time 60% while maintaining stable performance under load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Built **clean, reusable object-oriented API components** using Singleton and Factory patterns, AWS Glue, S3, SQL, and caching, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability.
- Automated **cross-platform builds, deployments, and 20+ unit/integration tests** with GitHub, Jenkins, PyTest, and Mockito, generating coverage reports and failure logs while supporting reliable releases across three operating systems.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **Node.js APIs across eight microservices** for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through WebSockets and Azure SQL while reducing cross-location consistency conflicts 33%.
- Implemented **secure role-based access and automated verification** with JWT, Jest, Selenium, Sentry, Docker, and Jenkins CI/CD, cutting deployment time 50% while improving reliability across billing and platform services.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **Spring Boot and Flask payment APIs** covering pricing, deposits, bookings, refunds, and status updates, maintaining sub-200-millisecond responses across 50,000+ records through Firebase synchronization and validated webhook processing.
- Implemented **multi-currency transaction and payout workflows** across cards, wallets, refunds, and disputes, reducing manual handling 70% while preserving consistent financial state through robust validation and idempotent webhook execution.

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, Bash
Backend: Spring Boot, FastAPI, Flask, Node.js, Express.js, REST APIs, microservices, WebSockets
Data & Transactions: PostgreSQL, MySQL, SQL Server, MongoDB, Firebase, data integrity, idempotency, webhooks
Computer Science: Data structures, algorithms, system design, OOP, design patterns, multithreading, distributed systems
Platform: AWS, Docker, Kubernetes, Git, GitHub Actions, Jenkins, CI/CD, PyTest, Mockito, Jest, Agile/Scrum

Projects

FinNexus Live | *Next.js, TypeScript, MongoDB, Kubernetes*

- Built a **financial platform supporting 1M+ transactions** with secure bank linking, real-time synchronization, advanced filtering, token-based access, MongoDB persistence, and Kubernetes-based CI/CD automation for 50,000+ users.

Chat2SQL Code | *Java, TypeScript, Oracle, Docker, Maven*

- Enhanced an **AI-powered database application** across Java and TypeScript components, supporting Oracle and 10+ SQL/NoSQL systems with Maven, Docker, schema exploration, query generation, and cross-platform debugging workflows.

CPU Scheduling Simulator Live | *C++, React.js, Data Structures, Algorithms*

- Implemented and tested **six scheduling algorithms** with React controls, Gantt visualization, edge-case handling, and dynamic metrics across 1,000+ process combinations, reducing average turnaround time 25% and debugging time 20%.